

Memory Matching Flutter Game

CS5450 Mobile Programming

Exercise 2

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Repository	https://github.com/luqi101/Mobile_Programming_Exercise_2.git
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Executive Overview

Memory Match is an original Flutter/Dart memory matching game created for CS5450 Exercise 2. The user selects a difficulty, flips cards to find matching pairs, tracks time and moves, and receives a final score on completion. The implementation uses local image assets, a responsive card grid, and controller-driven game logic.

Requirements Coverage

Requirement	Project coverage
Flutter/Dart implementation	Application source is implemented under lib/ using Flutter widgets and Dart models/controllers.
Memory matching functionality	Cards are generated in pairs, shuffled, flipped, matched, and reset after mismatches.
Configuration and run steps	README.md and this PDF include setup, Android, Chrome, analysis, and test commands.
Screenshots	Real screenshots from the screenshots/ folder are included.
Exact project structure	A generated project tree is included, excluding build/cache/IDE artifacts.
Local images/assets	Local PNG assets are stored in assets/images and registered in pubspec.yaml.
GitHub/ZIP readiness	The full Flutter project source, assets, tests, screenshots, and documentation are prepared for upload and p

Features

- Home screen with difficulty selection.
- Responsive game grid with animated card flip behavior.
- Two-card comparison with matched cards retained and mismatches flipped back after a delay.
- Input lock during mismatch delay to prevent rapid-tap errors.
- Timer, moves counter, matched-pair counter, restart, and new-game actions.
- Win/result screen with difficulty, time, moves, matched pairs, score, play again, and home actions.
- Original local card and background images generated for this project.

Game Rules And Difficulty

The player flips two cards at a time. Matching cards stay face up and count toward completion. Non-matching cards flip back after a short delay. A move is counted once per two-card attempt. The game finishes when every pair has been matched.

Difficulty	Pairs	Cards
Easy	4	8
Medium	6	12
Hard	8	16

Technical Implementation

- Game state is handled by GameController, a ChangeNotifier that owns deck generation, selected cards, move count, matched-pair count, timer cleanup, restart behavior, and win detection.

- Models are separated into Difficulty, MemoryCard, and GameResult.
- Scoring is isolated in lib/utils/scoring.dart and duration formatting is isolated in lib/utils/formatters.dart.
- Screens and reusable widgets are separated under lib/screens and lib/widgets.
- Assets are local PNG files under assets/images/cards and assets/images/backgrounds and are registered in pubspec.yaml.
- Tests cover controller behavior, matching/mismatch behavior, restart state, score calculation, and the core widget flow.

Setup And Run Instructions

```
flutter pub get
python3 tools/generate_assets.py
dart format .
flutter analyze
flutter test
flutter devices
flutter run -d <android-device-id>
flutter run -d chrome
```

For Android testing, start an emulator or connect a physical device, then replace with the id shown by flutter devices. Chrome/Web is also supported.

Testing And Verification

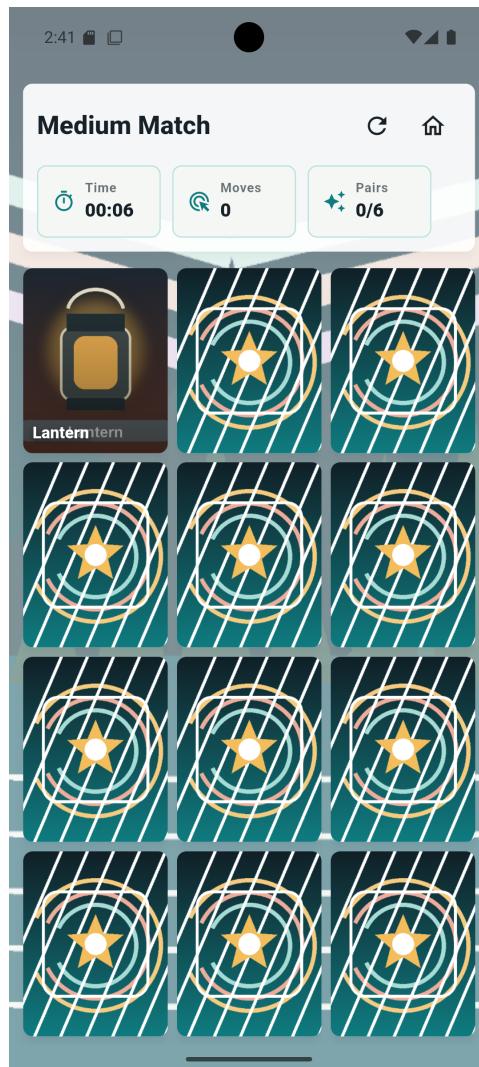
Command	Latest result
flutter clean	Pass
flutter pub get	Pass
dart format .	Pass, no changes required
flutter analyze	Pass, no issues found
flutter test	Pass, 13 tests
flutter devices	Android emulator and Chrome detected

Screenshots

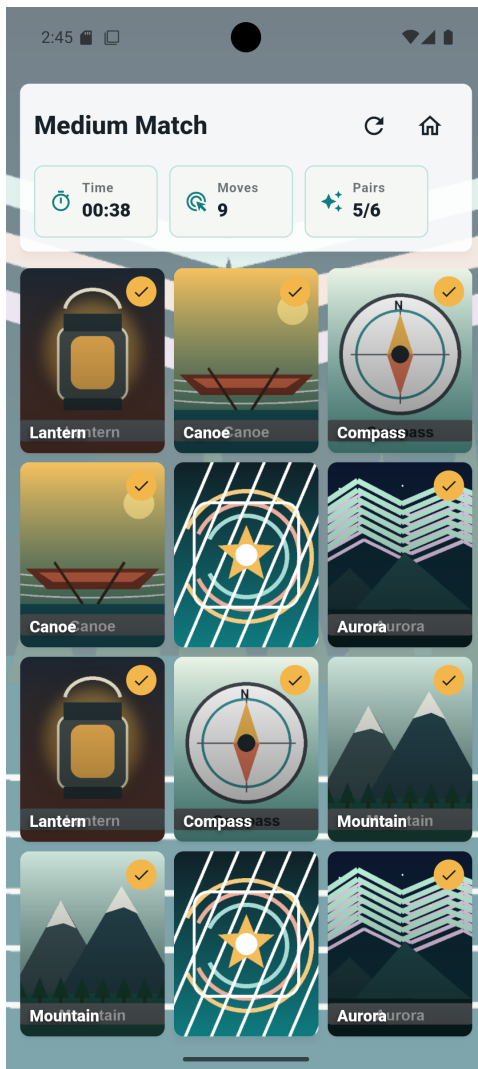
Home screen and difficulty selection



Active game screen



Near-complete game state



Win/result screen



Exact Project Structure

The following structure is generated from the current project directory. Build outputs, cache folders, Git metadata, IDE caches, and temporary OS files are excluded.

```
|-- android/
|   |-- .kotlin/
|   |   |-- sessions/
|   |-- app/
|   |   |-- src/
|   |   |   |-- debug/
|   |   |   |   |-- AndroidManifest.xml
|   |   |   |-- main/
|   |   |   |   |-- java/
|   |   |   |   |   |-- io/
|   |   |   |   |   |   |-- flutter/
|   |   |   |   |   |   |-- plugins/
|   |   |   |   |-- kotlin/
|   |   |   |   |   |-- com/
|   |   |   |   |   |   |-- luqmanaadil/
|   |   |   |   |   |   |   |-- memory_matching_flutter_game/
|   |   |   |   |   |   |   |-- MainActivity.kt
|   |   |   |-- res/
|   |   |   |   |-- drawable/
|   |   |   |   |   |-- launch_background.xml
|   |   |   |   |-- drawable-v21/
|   |   |   |   |   |-- launch_background.xml
|   |   |   |   |-- mipmap-hdpi/
|   |   |   |   |   |-- ic_launcher.png
|   |   |   |   |-- mipmap-mdpi/
|   |   |   |   |   |-- ic_launcher.png
|   |   |   |   |-- mipmap-xhdpi/
|   |   |   |   |   |-- ic_launcher.png
|   |   |   |   |-- mipmap-xxhdpi/
|   |   |   |   |   |-- ic_launcher.png
|   |   |   |   |-- mipmap-xxxhdpi/
|   |   |   |   |   |-- ic_launcher.png
|   |   |   |   |-- values/
|   |   |   |   |   |-- styles.xml
|   |   |   |   |-- values-night/
|   |   |   |   |   |-- styles.xml
|   |   |   |-- AndroidManifest.xml
|   |   |-- profile/
|   |   |   |-- AndroidManifest.xml
|   |-- build.gradle.kts
|-- gradle/
|   |-- wrapper/
|   |   |-- gradle-wrapper.jar
|   |   |-- gradle-wrapper.properties
|-- .gitignore
|-- build.gradle.kts
|-- gradle.properties
|-- gradlew
|-- gradlew.bat
|-- settings.gradle.kts
|-- assets/
|   |-- images/
|   |   |-- backgrounds/
|   |   |   |-- game_background.png
|   |   |   |-- home_background.png
|   |   |-- cards/
|   |   |   |-- card_aurora.png
|   |   |   |-- card_back.png
|   |   |   |-- card_canoë.png
|   |   |   |-- card_compass.png
|   |   |   |-- card_lake.png
|   |   |   |-- card_lantern.png
|   |   |   |-- card_mountain.png
|   |   |   |-- card_pine.png
|   |   |   |-- card_star.png
|-- ios/
|   |-- Flutter/
|   |   |-- AppFrameworkInfo.plist
|   |   |-- Debug.xcconfig
|   |   |-- Release.xcconfig
|   |-- Runner/
|   |   |-- Assets.xcassets/
|   |   |   |-- AppIcon.appiconset/
|   |   |   |   |-- Contents.json
|   |   |   |   |-- Icon-App-1024x1024@1x.png
|   |   |   |   |-- Icon-App-20x20@1x.png
|   |   |   |   |-- Icon-App-20x20@2x.png
```

```

|-- Icon-App-20x20@3x.png
|-- Icon-App-29x29@1x.png
|-- Icon-App-29x29@2x.png
|-- Icon-App-29x29@3x.png
|-- Icon-App-40x40@1x.png
|-- Icon-App-40x40@2x.png
|-- Icon-App-40x40@3x.png
|-- Icon-App-60x60@2x.png
|-- Icon-App-60x60@3x.png
|-- Icon-App-76x76@1x.png
|-- Icon-App-76x76@2x.png
|-- Icon-App-83.5x83.5@2x.png
|-- LaunchImage.imageset/
|-- Contents.json
|-- LaunchImage.png
|-- LaunchImage@2x.png
|-- LaunchImage@3x.png
|-- README.md
|-- Base.lproj/
|-- LaunchScreen.storyboard
|-- Main.storyboard
|-- AppDelegate.swift
|-- GeneratedPluginRegistrant.h
|-- GeneratedPluginRegistrant.m
|-- Info.plist
|-- Runner-Bridging-Header.h
|-- SceneDelegate.swift
|-- Runner.xcodeproj/
|-- project.xcworkspace/
|-- xcshareddata/
|-- IDEWorkspaceChecks.plist
|-- WorkspaceSettings.xcsettings
|-- contents.xcworkspacedata
|-- xcshareddata/
|-- xcschemes/
|-- Runner.xcscheme
|-- project.pbxproj
|-- Runner.xcworkspace/
|-- xcshareddata/
|-- IDEWorkspaceChecks.plist
|-- WorkspaceSettings.xcsettings
|-- contents.xcworkspacedata
|-- RunnerTests/
|-- RunnerTests.swift
|-- .gitignore
-- lib/
|-- app/
|-- app_theme.dart
|-- memory_match_app.dart
|-- controllers/
|-- game_controller.dart
|-- models/
|-- difficulty.dart
|-- game_result.dart
|-- memory_card.dart
|-- screens/
|-- game_screen.dart
|-- home_screen.dart
|-- result_screen.dart
|-- utils/
|-- formatters.dart
|-- scoring.dart
|-- widgets/
|-- asset_background.dart
|-- difficulty_selector.dart
|-- game_header.dart
|-- memory_card_tile.dart
|-- primary_action_button.dart
|-- responsive_game_grid.dart
|-- main.dart
-- screenshots/
|-- Screenshot_20260520_024200.png
|-- Screenshot_20260520_024216.png
|-- Screenshot_20260520_024524.png
|-- Screenshot_20260520_024535.png
-- test/
|-- game_controller_test.dart
|-- widget_test.dart
-- tools/
|-- build_readme_pdf.py
|-- generate_assets.py
-- web/
|-- icons/
|-- Icon-192.png
|-- Icon-512.png
|-- Icon-maskable-192.png

```

```
| | `-- Icon-maskable-512.png
| |-- favicon.png
| |-- index.html
| |-- manifest.json
|-- .gitignore
|-- .metadata
|-- AGENTS.md
|-- analysis_options.yaml
|-- pubspec.lock
|-- pubspec.yaml
|-- README.md
`-- README.pdf
```

Final Submission Checklist

Item	Status
Dart source files present	Pass
Local assets/images present	Pass
Real screenshots present	Pass
README.md present	Pass
README.pdf present	Pass
flutter analyze	Pass
flutter test	Pass
GitHub repository	Prepared for public push to the repository URL on the cover page
ZIP/D2L package	Ready to package after final repository verification